

Quantum Riesz Transform Boundedness via Quantum Euclidean Singular Integrals

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Status

Literature-implied answer. The Riesz transform conjecture in Lafleche’s *On Quantum Sobolev Inequalities* follows from the Calderon–Zygmund theory for quantum Euclidean spaces of Gonzalez-Perez, Junge, and Parcet, after identifying the Weyl phase-space algebra with the Heisenberg–Weyl instance of $\mathcal{R}_\Theta$.

Source Question

The source paper is Laurent Lafleche, *On Quantum Sobolev Inequalities*, arXiv:2210.03013, J. Funct. Anal. 286 (2024), 110400. In the subsection ”Boundedness of Riesz transforms in the quantum phase space”, the paper proves boundedness for the split Schur multipliers $(-\Delta_{\hbar,\xi})^{-1/2}D_{\hbar,\xi}$ and, by Fourier conjugation, $(-\Delta_{\hbar,x})^{-1/2}D_{\hbar,x}$. It then states that it is not clear whether these split expressions are equivalent to

$$\|(-\Delta_{\hbar})^{1/2}A\|_{L^p}$$

when $p \neq 2$, and conjectures more precisely that

$$(-\Delta_{\hbar})^{-1/2}D_{\hbar} : L^p \longrightarrow (L^p)^{2d}$$

is bounded for every $1 < p < \infty$, with norm independent of $\hbar$.

Supporting Theorem

The supporting paper is A. M. Gonzalez-Perez, M. Junge, and J. Parcet, *Singular integrals in quantum Euclidean spaces*, arXiv:1705.01081, Memoirs Amer. Math. Soc. 272 (2021), no. 1334. It defines quantum Euclidean spaces $\mathcal{R}_\Theta$, notes that the choice

$$\Theta = 2\pi\hbar I_m \otimes \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

yields the Heisenberg–Weyl algebra, and proves quantum Calderon–Zygmund extrapolation: if a kernel operator on $\mathcal{R}_\Theta$ has the L_2 cancellation estimate and the required quantum kernel smoothness, then it is completely bounded on $L_p(\mathcal{R}_\Theta)$ for all $1 < p < \infty$. The paper also emphasizes that the convolution-kernel case, equivalently quantum Fourier multipliers, is easier and is covered by the same framework.

Implication

Let $n = 2d$. Under Weyl quantization in Lafleche's notation,

$$D_{\hbar,j} \text{op}_f = \text{op}_{\partial_{z_j} f}, \quad (-\Delta_{\hbar})^s \text{op}_f = \text{op}_{(-\Delta_z)^s f}.$$

Thus on the Fourier side in the phase-space variable z , the j -th component

$$R_{\hbar,j} = D_{\hbar,j}(-\Delta_{\hbar})^{-1/2}$$

has multiplier

$$m_j(\eta) = c \frac{i\eta_j}{|\eta|}, \quad \eta \neq 0,$$

up to the harmless Fourier-normalization constant c . This is exactly the classical Riesz transform symbol, now acting as a scalar quantum Fourier multiplier on the quantum Euclidean space $\mathcal{R}_{\Theta_{\hbar}}$.

Equivalently, its principal-value kernel is transported from the classical Calderon–Zygmund kernel

$$K_j(z) = c_n \frac{z_j}{|z|^{n+1}}.$$

The normal homomorphism π_{Θ} used in the supporting paper transports the Euclidean distance-to-the-diagonal and its differential estimates into $\mathcal{R}_{\Theta}$. Consequently K_j satisfies the supporting paper's quantum Calderon–Zygmund hypotheses with constants depending only on n and the normalization of K_j , not on the deformation matrix Θ . The L_2 norm is the Plancherel multiplier norm $\|m_j\|_{\infty}$, also independent of Θ . Hence for each $1 < p < \infty$,

$$\|R_{\hbar,j}A\|_{L^p} \leq C_{p,n}\|A\|_{L^p}, \quad j = 1, \dots, n,$$

with $C_{p,n}$ independent of $\hbar$.

Vector-Valued Bound

The source appendix gives the needed finite-dimensional vector comparison for operator columns. If $B = (B_1, \dots, B_n)$, then for $p \geq 2$,

$$\|B\|_{S^p(\ell^2)} \leq \left(\sum_{j=1}^n \|B_j\|_{S^p}^2 \right)^{1/2},$$

whereas for $1 \leq p \leq 2$,

$$\|B\|_{S^p(\ell^2)} \leq \left(\sum_{j=1}^n \|B_j\|_{S^p}^p \right)^{1/p}.$$

Applying these inequalities to $B_j = R_{\hbar,j}A$ gives

$$\|(-\Delta_{\hbar})^{-1/2}D_{\hbar}A\|_{(L^p)^n} \leq C'_{p,n}\|A\|_{L^p},$$

where $C'_{p,n}$ is independent of $\hbar$. This is the conjectured boundedness.

The same scalar estimates also give the reverse norm comparison. Indeed, on the dense Weyl-Schwartz core,

$$R_{\hbar,j} = D_{\hbar,j}(-\Delta_{\hbar})^{-1/2},$$

and the Fourier multiplier identity

$$(-\Delta_{\hbar})^{1/2}A = -\sum_{j=1}^n R_{\hbar,j} D_{\hbar,j} A$$

holds on the dense Schwartz/Weyl core and extends by density. Therefore

$$\|(-\Delta_{\hbar})^{1/2}A\|_{L^p} \leq C''_{p,n} \|D_{\hbar}A\|_{(L^p)^n},$$

again with no $\hbar$ -dependence. Thus the first-order quantum Bessel and Sobolev seminorms are equivalent for fixed d and $1 < p < \infty$, with constants independent of $\hbar$.

Scope and Search

This is a literature-implied answer, not a new theorem. The supporting paper predates the source question and does not appear to explicitly state that it answers Lafleche’s conjecture. The implication uses the standard identification of the source’s Weyl/Schatten phase-space model with the Heisenberg–Weyl quantum Euclidean space. Constants are allowed to depend on p and d ; no dimension-free assertion is claimed.

Local run indexes were searched for arXiv:2210.03013, the title, quantum Sobolev phrases, Riesz transform phrases, and Junge–Parcet/Gonzalez-Perez keywords. Web searches on 2026-07-18 checked arXiv and EMS pages for arXiv:2210.03013, arXiv:1705.01081, arXiv:1407.2475, and close phrase searches. No existing run packet or explicit later answer to this exact conjecture was found.

References

- [1] L. Lafleche. *On Quantum Sobolev Inequalities*. J. Funct. Anal. 286 (2024), 110400. arXiv:2210.03013.
- [2] A. M. Gonzalez-Perez, M. Junge, and J. Parcet. *Singular integrals in quantum Euclidean spaces*. Memoirs Amer. Math. Soc. 272 (2021), no. 1334. arXiv:1705.01081.
- [3] M. Junge, T. Mei, and J. Parcet. *Noncommutative Riesz transforms – dimension free bounds and Fourier multipliers*. J. Eur. Math. Soc. 20 (2018), 529–595. arXiv:1407.2475.